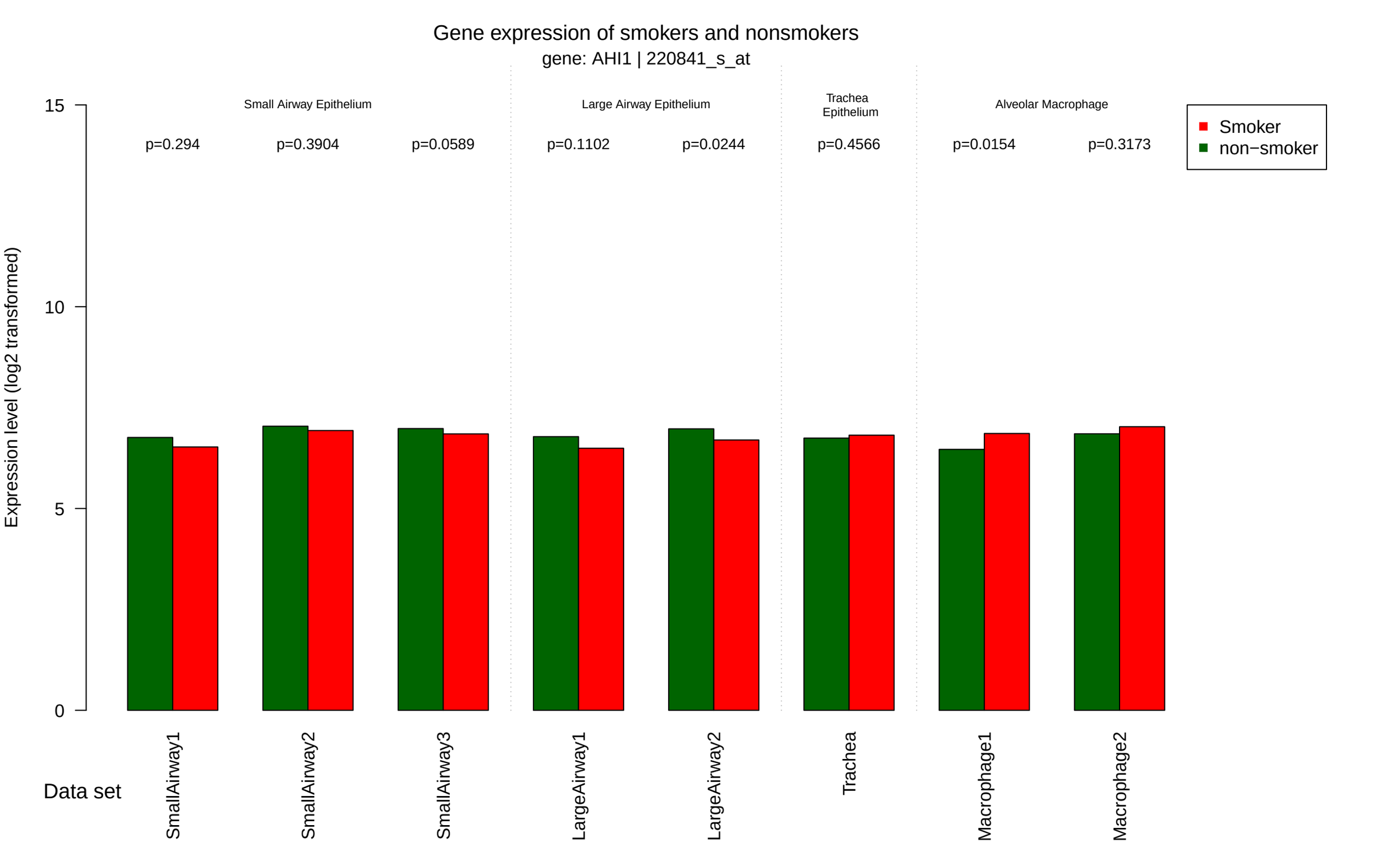
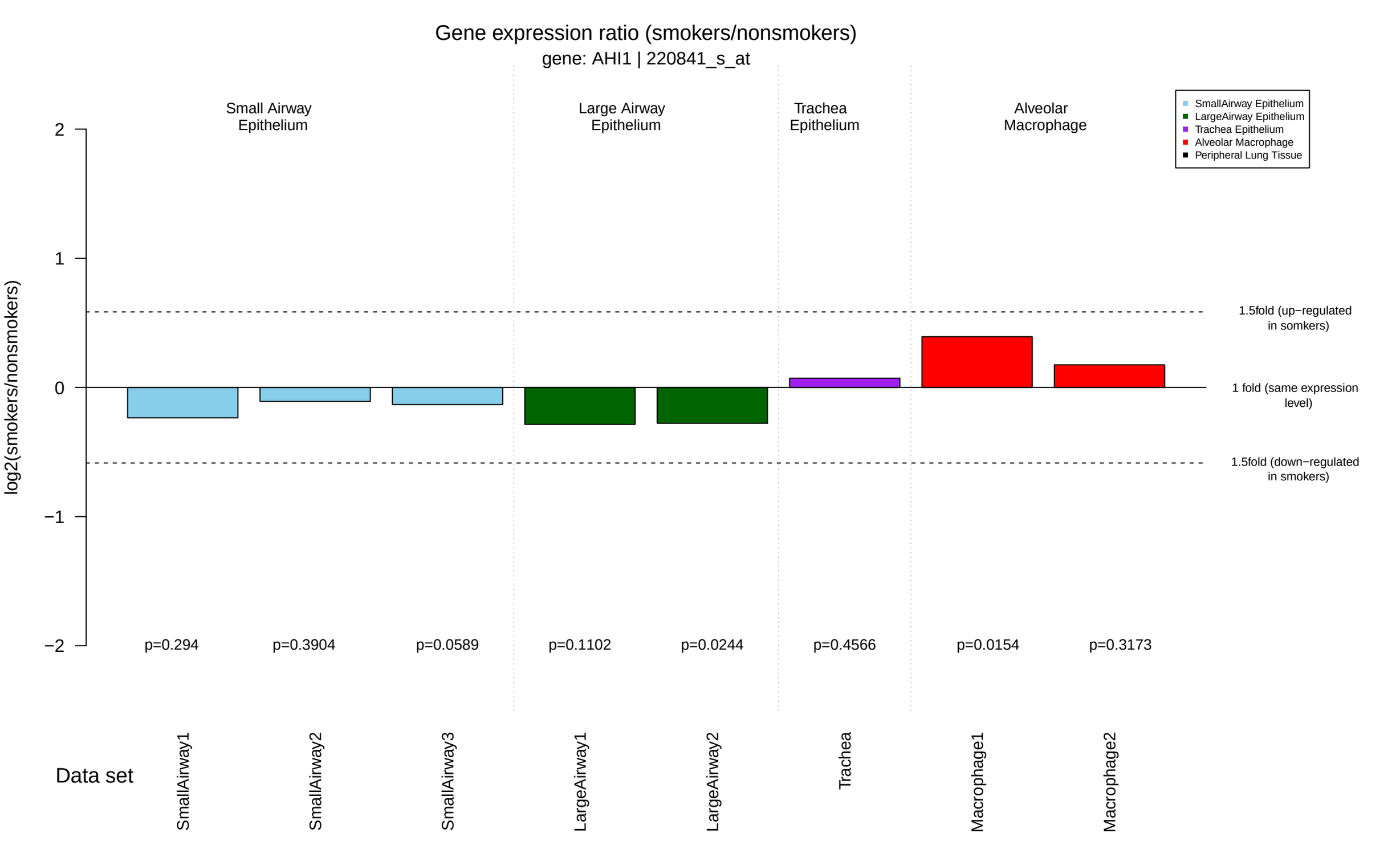
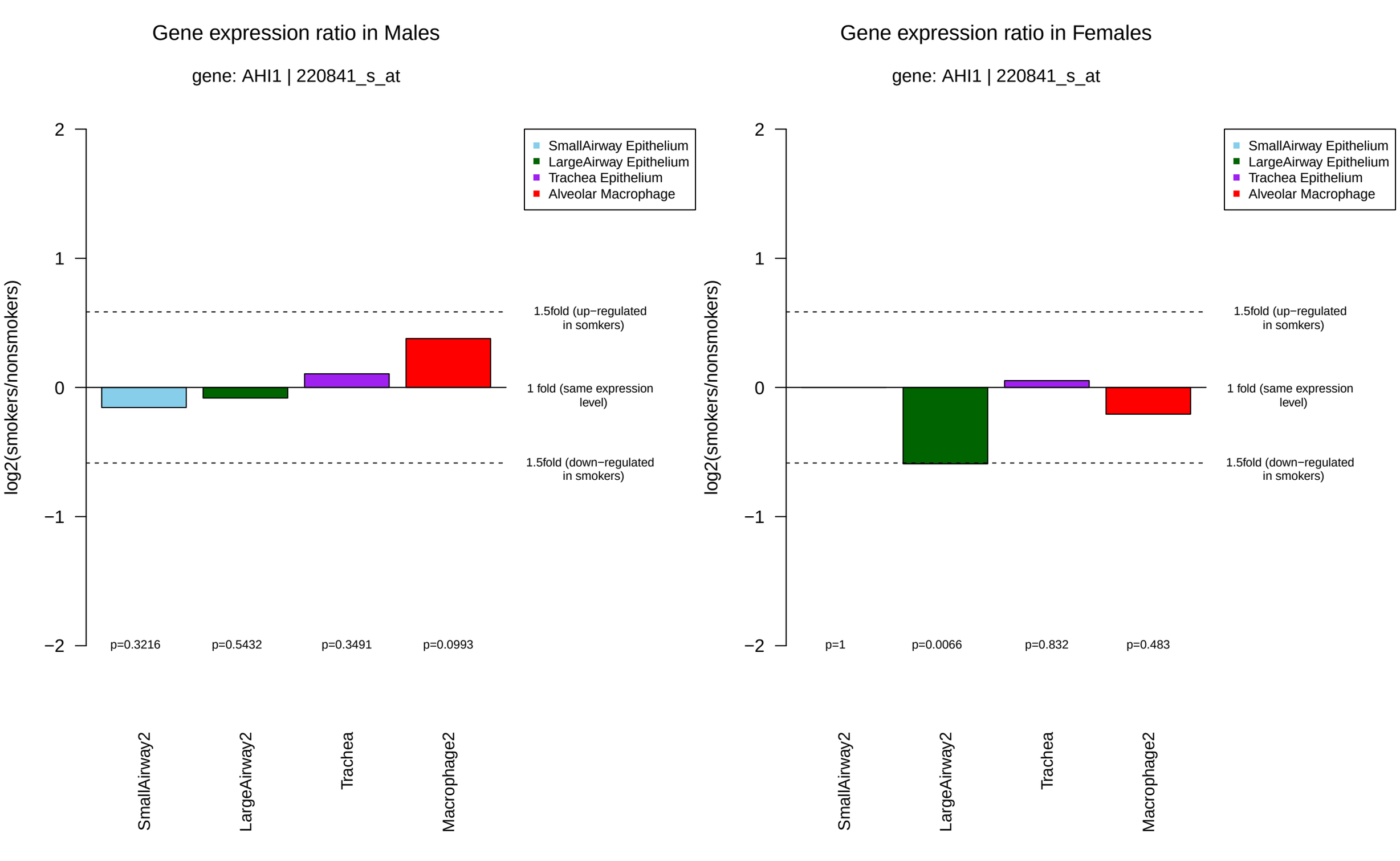
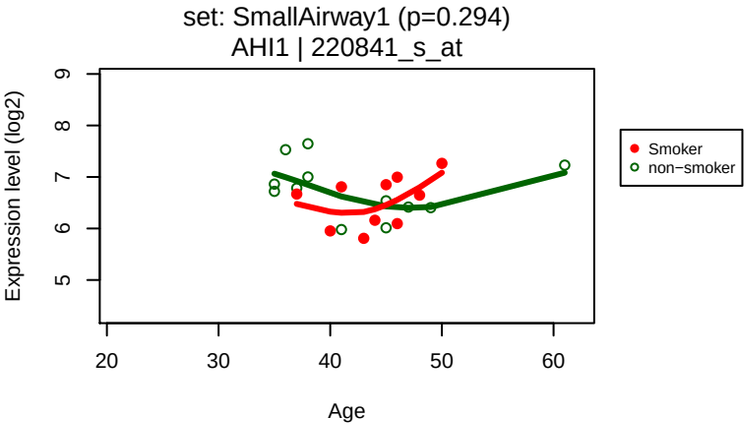
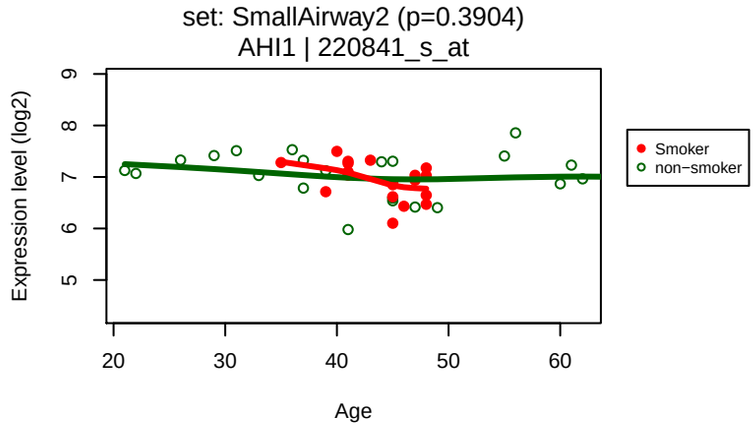
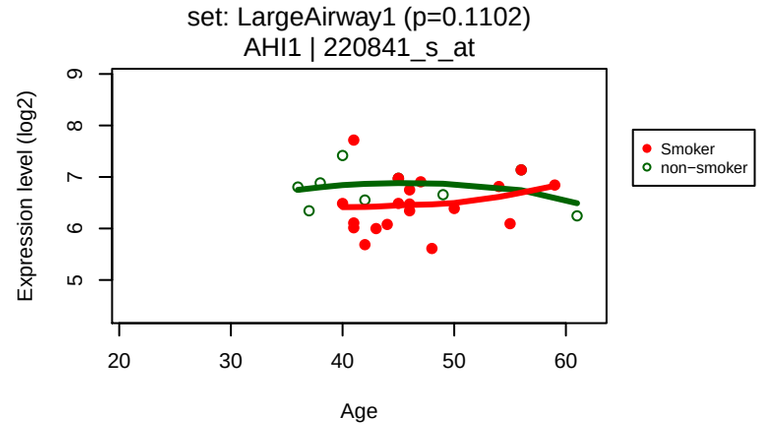

Gene expression of smokers and nonsmokers
gene: AHI1 | 220841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.294
p=0.3904
p=0.0589
p=0.1102
p=0.0244
p=0.4566
p=0.0154
p=0.3173
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

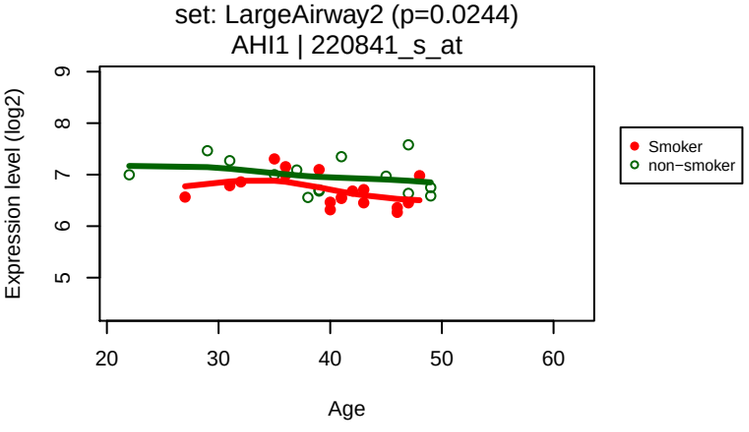

Gene expression ratio (smokers/nonsmokers)
gene: AHI1 | 220841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.294
p=0.3904
p=0.0589
p=0.1102
p=0.0244
p=0.4566
p=0.0154
p=0.3173
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

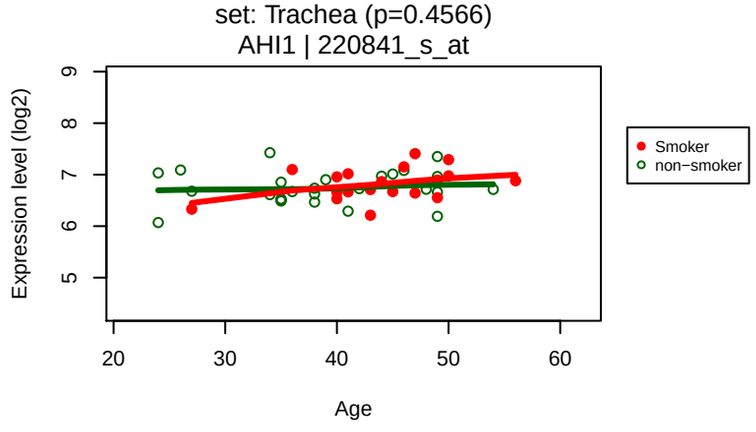

Gene expression ratio in Males
gene: AHI1 | 220841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3216
p=0.5432
p=0.3491
p=0.0993
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AHI1 | 220841_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1
p=0.0066
p=0.832
p=0.483
SmallAirway2
LargeAirway2
Trachea
Macrophage2

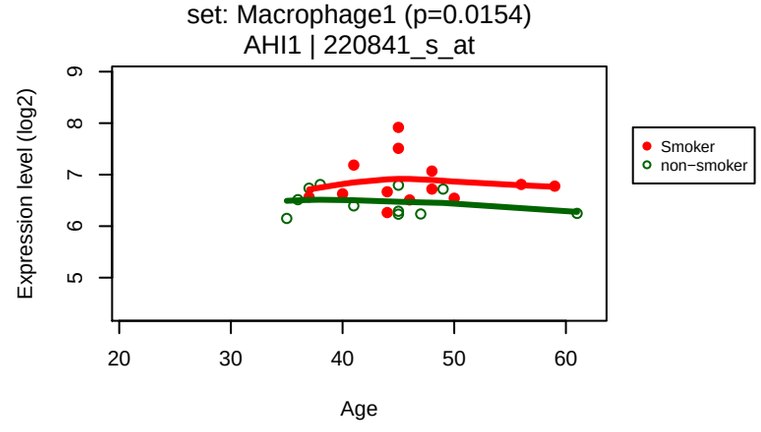

set: SmallAirway1 (p=0.294)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker

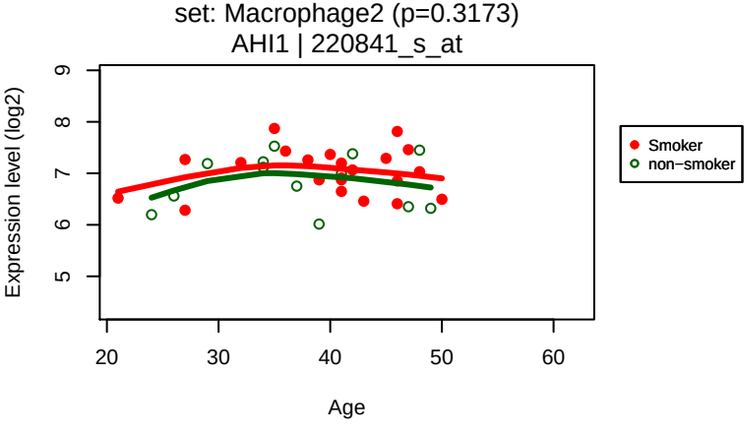

set: SmallAirway2 (p=0.3904)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker

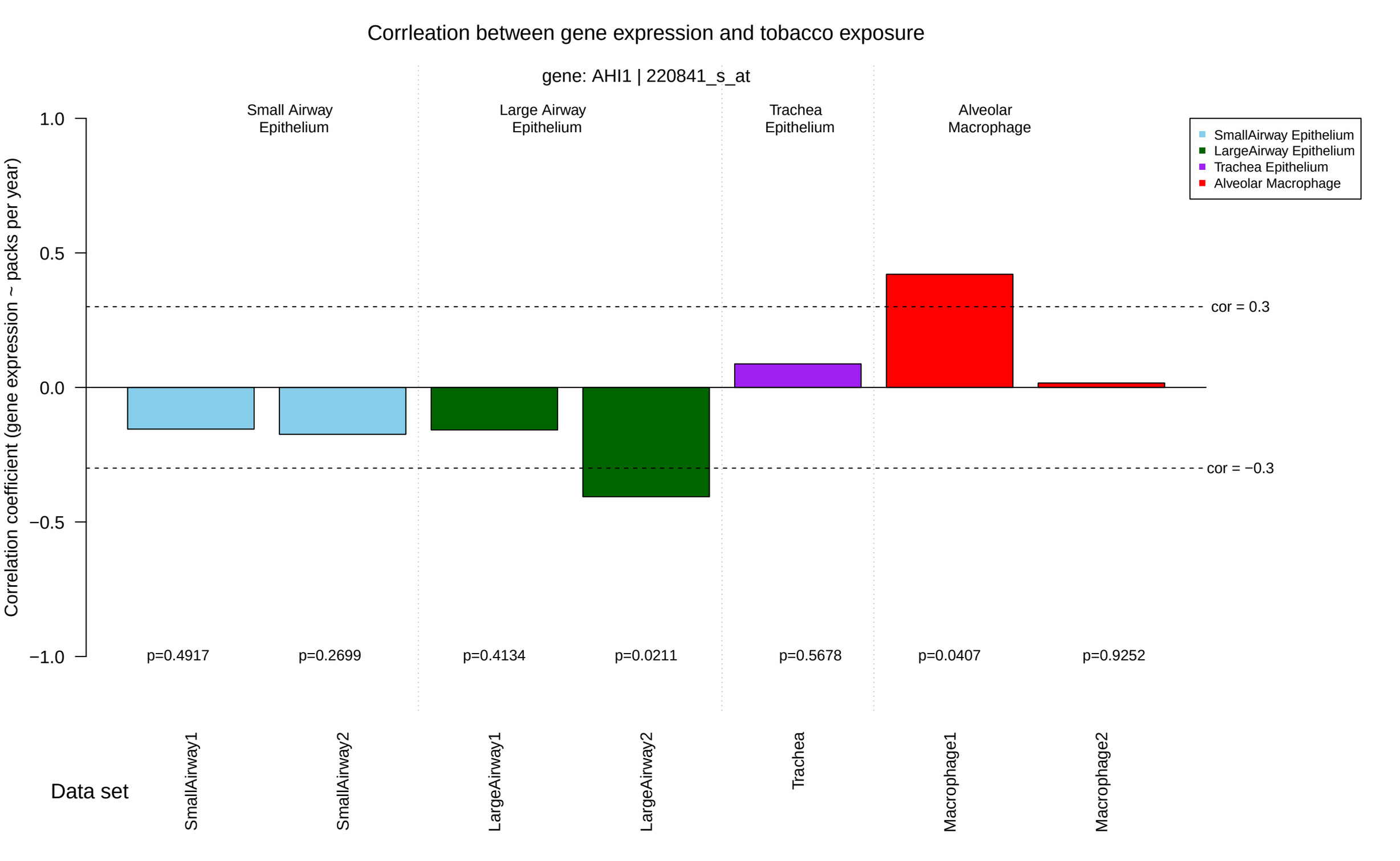

set: LargeAirway1 (p=0.1102)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0244)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4566)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0154)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3173)
AHI1 | 220841_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AHI1 | 220841_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4917
p=0.2699
p=0.4134
p=0.0211
p=0.5678
p=0.0407
p=0.9252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2